

Cosc 30: Discrete Mathematics

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Problem 1.

- (a) Prove that any graph G with n vertices and n edges must have a vertex v with degree at most 2.
- (b) Draw a simple graph that has 10 edges, 10 vertices, with no vertex with degree exactly 2. What is the largest number of degree (≤ 1) vertices you can have?

Solution.

- (a) Let V be the set of all vertices, note that $|V| = n$. Assume for the sake of contradiction that there exist no nodes with degree ≤ 2 . Then the minimum degree any node can have is 3. That is

$$\sum_{v \in V} \deg(v) \geq 3n$$

But by the Handshake Lemma, we know,

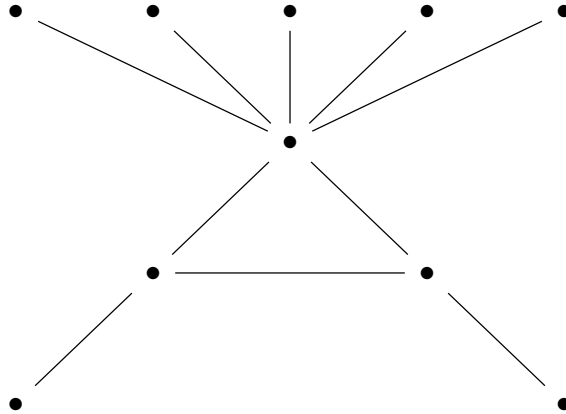
$$\sum_{v \in V} \deg(v) = 2n$$

That is,

$$\sum_{v \in V} \deg(v) = 2n \geq 3n$$

A contradiction! That is, any graph G with n vertices and n edges must have at least a vertex v with degree at most 2.

(b) 7 is the largest number of degree (≤ 1) vertices we can have.



Note there cannot exist 8 or more such vertices because by the handshake lemma we know, that the sum of the degrees of the vertices must add up to 20. Suppose $k = 8$ vertices have degree ≤ 1 . They contribute at most 8 to the degree sum, so the remaining 2 vertices must contribute at least $20 - 8 = 12$, i.e. their degrees sum to at least 12. Now, since these two nodes can only have 1 edge between them (contributing 2 to the degree sum), adding more edges would mean connecting the nodes to one of the other 8 nodes and hence bumping up that nodes' degree by 1.

That is, 7 is the maximum number of degree (≤ 1) vertices we can have.

Problem 2.

- (a) Prove that any graph containing odd-length closed walks must also contain odd cycles.
- (b) Draw a graph that contains even-length closed walks but has no even cycles.

Solution.

- (a) Let W be an odd-length closed walk in a graph G , and let W' be the shortest odd-length walk in G . Then we claim that W' is an odd-cycle.

We will show this by contradiction. Assume that W' is not a cycle, then there is some vertex v that W' visits twice. Then we can break W' up into two closed walks W_1, W_2 at v , such that,

$$|W_1| + |W_2| = |W'|$$

Since $|W'|$ is odd, one of $|W_1|$ or $|W_2|$ is odd, and the other even. But then the odd-length closed walk is strictly shorter than W' , contradicting minimality!

That is, W' must be a cycle.

- (b) Note that the graph below has an even-length closed walk, namely,

$$A \rightarrow B \rightarrow C \rightarrow A \rightarrow B \rightarrow C \rightarrow A$$

but only one cycle (which has odd-length),

$$A \rightarrow B \rightarrow C \rightarrow A$$

